

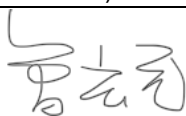
FACULTY OF ARTS & SOCIAL SCIENCES
HEADS & DEANERY RESEARCH SUPPORT SCHEME (HDRSS) 2026

Section 1	
Name of Applicant:	CHAN Kwang Guan Daniel
Department:	Centre for Language Studies
Appointment:	Asst Dean, Undergraduate Matters
Period of Headship, Asst. or Vice- Deanship:	Up to 30 June 2027

Section 2		
(A) Title of Research: Guided AI-powered Language Learning via Augmented & eXtended Interactional Mediations (GALAXIM)		
(B) Research agenda: This project explores how human and technological mediation can work together in AI-powered beginner French learning — augmenting situated, reflective learner interaction, extending opportunities for practice, and generating design principles for human-centered AI/XR-supported learning environments. Building on existing proof-of-concept platforms and custom tools, the project refines these through AI-assisted rapid prototyping, developing learner-facing interfaces, task flows, feedback screens, and trial activities. The pilot centres on two A1-level communicative scenarios, with culturally situated language use serving as an intercultural-mediation lens across both. AI-mediated feedback and scaffolding form the spine of the design; in one scenario, an augmented/extended-reality (XR) probe is added to explore situated, immersive task performance. Each scenario follows the same cycle: pre-task preparation, AI-supported (and, in the probe, XR-supported) task performance, peer- or teacher-guided discussion, and post-task reflection. To consolidate each scenario's target structures, the environment also incorporates short, AI-generated, learner-tailored songs that embed them, exploiting music's mnemonic and prosodic properties to aid recall; all lyrics are vetted for grammatical and prosodic accuracy before use. The study adopts a design-based, exploratory pilot methodology — a single design–enact–reflect iteration — with roughly 30 beginner learner-participants in the classroom and a smaller subset taking part in focus groups or more intensive feedback sessions, subject to enrolment, timetable feasibility, and ethics approval. Data comprise learner-experience feedback, task responses, observation notes, focus-group data, and, where appropriate, audio recordings of learner production. Analysis centres on the interplay between human and technological mediation and its effects on learner engagement and oral production; confidence, written production, intercultural mediation, and perceived learning value are reported as secondary, largely descriptive dimensions. Three research questions guide the project: (1) How do beginner French learners experience human and technological mediation in GALAXIM-supported tasks? (2) How do AI-supported activities — and, in one scenario, an XR-supported activity — shape learner engagement, confidence, and language production? (3) What design principles for future human-centred AI/XR-mediated platforms, and what redesign recommendations, emerge from this first iteration?		
(C) Expected outcomes: (1) A small set of reusable mediated learning tasks and classroom activity sequences combining teacher/peer mediation with AI- and XR-supported interaction; (2) Pilot evidence on learner engagement, confidence, usability, intercultural noticing, oral/written production and perceived learning value; (3) A set of design principles for future AI-assisted rapid prototyping and customized language-learning platform development.		
(D) Research Budget:		
Category	Amount requested (\$)	Description of cost items and justification

(i) Manpower	S\$5,800.00	RA support for prototype build and content authoring, pilot logistics, transcription and coding of learner oral production (wav2vec2/IPA pipeline), and preliminary analysis. 140 UG hours @ S\$20/hr (S\$2,800) + 100 graduate hours @ S\$30/hr (S\$3,000), at NWS rates (to be confirmed).
SUB-TOTAL	S\$5,800.00	
(ii) Equipment [Only small equipment is supported. Laptops and computer equipment are not eligible.] Please provide quotation.	S\$2,000.00	Three dedicated research items (quotations to follow). (1) Meta Quest 3 512GB headset (~S\$900): instrument for the immersive/augmented mediation condition under study; one shared unit, inventoried and returned on completion, pending Research Division eligibility confirmation. (2) 360 camera (~S\$780): captures immersive scenes for AR/XR vocabulary labelling in the A1 scenarios. (3) Audio recorder and mic (~S\$320): captures learner oral production and focus groups. Sources: Challenger (online.challenger.sg/products/meta-quest-3-512gb); Cathay Photo (cathayphoto.com.sg/cameras/vr-360-cameras/insta360-x5); Swee Lee (sweelee.com.sg/products/zoom-h1essential-handy-recorder).
SUB-TOTAL	S\$2,000.00	
(iii) OOE - Overseas Travel -- Conference attendance for paper presentation (Covers economy airfare, accommodation and per diem)		
Return airfare	S\$500.00	23rd AsiaCALL Conference, 14–15 Nov 2026; Phuong Dong University, Hanoi (NUS Sem 1 reading week).
Accommodation 3 nights	S\$450.00	Source: asiacall.info . Per diem per NUS Annex B; airfare/accommodation to confirm at booking.
Per diem and local transport	S\$650.00	
Conference registration	S\$400.00	
Economy return airfare	S\$1,600.00	2026 Joint ALAA-ALANZ-ALTAANZ Conference, 26–28 Nov 2026; University of Queensland.
Accommodation 4 nights	S\$1,600.00	Source: alaa.net.au/2026 . Per diem per NUS Annex B; airfare/accommodation to confirm at booking.
Per diem and local transport	S\$1,300.00	
Conference registration	S\$700.00	
SUB-TOTAL	S\$7,200.00	
(iv) OOE – Consumables/ Training/ Other Miscellaneous		
Fieldwork: classroom-pilot learner tokens, 30 users x S\$20	S\$600.00	Token amounts follow NUS-IRB participant-reimbursement norms and the approved protocol. Source: nus.edu.sg/irb .
Fieldwork: focus-group tokens, 2 sessions x 6 users x S\$25	S\$300.00	
Fieldwork: additional usability or make-up tokens, 10 users x S\$20	S\$200.00	
Fieldwork: focus-group refreshments and materials	S\$300.00	
Fieldwork: additional participants or approved token adjustments	S\$300.00	

SaaS/Software: AI agent and model subscriptions for dialogue prototyping, feedback testing, prompt comparison and AI-assisted rapid prototyping	S\$800.00	Core technological mediation under study, and the prototyping engine for task flows and feedback. Source: anthropic.com/pricing ; openai.com/api/pricing .
SaaS/Software: Subscription to Suno AI (Pro) for AI-generated, learner-tailored songs embedding target A1 structures	S\$160.00	AI song generation (Suno Pro, ~US\$10/mo) for learner-tailored songs embedding A1 structures as mnemonic support; lyrics vetted before use. Source: suno.com/pricing .
SaaS/Software: AI coding/dev assistant (Claude Pro / Cursor / Copilot), ~12 months	S\$200.00	AI coding assistant to speed prototype refinement (interfaces, task flows, feedback screens). Source: github.com/features/copilot/plans ; cursor.com/pricing ; anthropic.com/pricing .
SaaS/Software: Cloud compute/hosting coverage (Firebase Blaze, pay-as-you-go) + domain	S\$120.00	Hosts the prototype for classroom pilots beyond free-tier limits. Source: firebase.google.com/pricing .
SaaS/Software: Research consumables & materials (printing of task/feedback materials; storage media for recordings)	S\$200.00	Printing of task/feedback materials, and high-endurance microSD for recordings. Source: gaincity.com .
SUB-TOTAL	S\$3,180.00	
Total amount requested (i + ii + iii + iv)	S\$18,180.00	



10 June 2026

Signature of applicant and Date

Section 3 : To be completed by Dean or Nominee

(i) Application is supported/not supported* (* please delete accordingly)

(ii) Comments:

Signature: _____

Date: _____

Note:**Please submit application to the Research Division (Attn: Ms Julie Ng)**